

Benchmarking unsupervised anomaly detection in brain MRI

*A systematic comparison across training regimes
and preprocessing pipelines*

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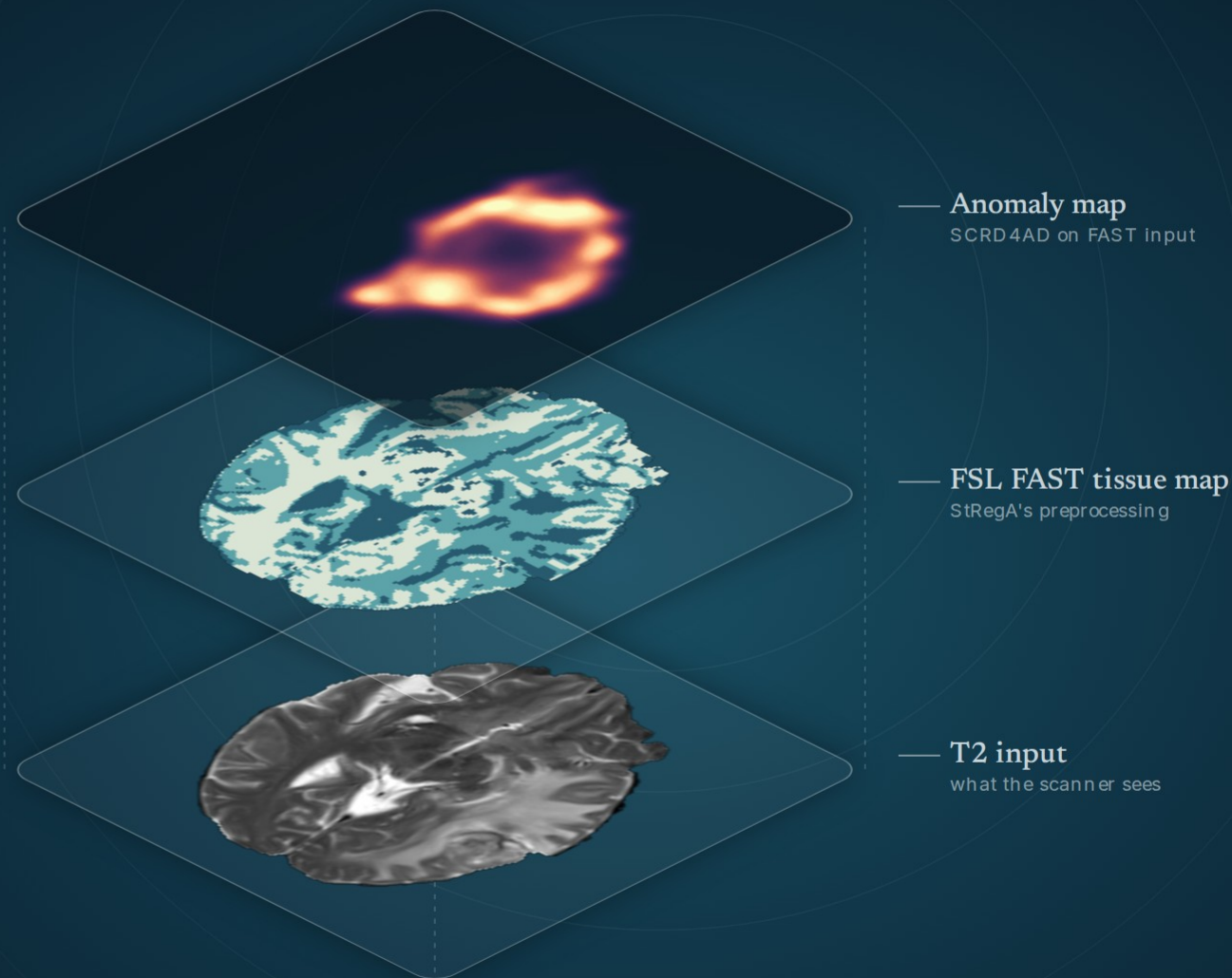
UAD METHODS

10

CONFIGURATIONS EACH

62

EVALUATED RUNS



BEFORE WE START

Declarations

Conflicts of interest

No conflict of interest to disclose.

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Data: IXI, MOOD and BraTS 2020, all publicly available.

THE QUESTION

Is the gain the model, or the pipeline around it?

Diffusion models report big gains, but every study trains on different data, contrasts and preprocessing. We ran five methods through one shared protocol.



1 Healthy scans

IXI 3T and 1.5T
T1 and T2
MOOD 3T, T1
10 configurations



2 What we varied

Contrast
Pooling
Preprocessing
compared in turn



3 Five methods

- StRegA
- SCRD4AD
- cDDPM
- MAD-AD
- UCCD



4 Tumour test set

BraTS 2020
T1 and T2
one metric code



5 Metrics

Slice DICE
Volume DICE
Slice AUROC
by tumour size

5 UAD methods · 10 matched configurations each · 1 metric code · 62 runs

Contrast and training data move the results

CONTRAST

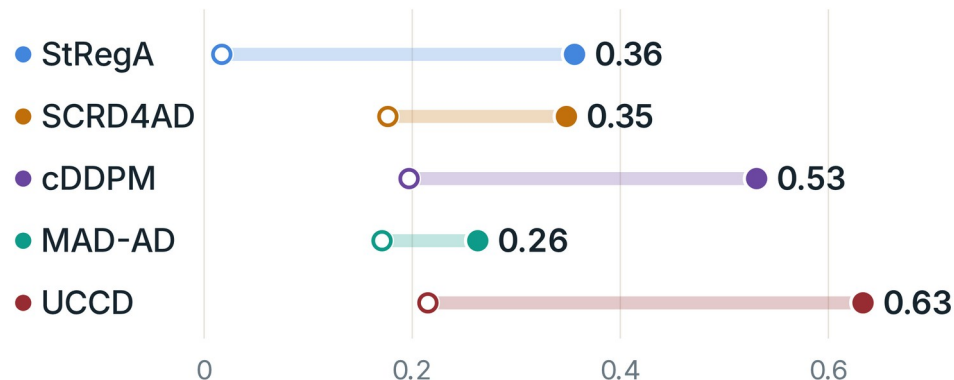
Contrast matters as much as the model

+0.42

best Slice DICE gained by UCCD, trained and tested on T2 instead of T1

Best Slice DICE, trained and tested on

○ T1 ● T2



Every method segmented best on T2. UCCD's gain is larger than the 0.37 gap between our best and worst method.

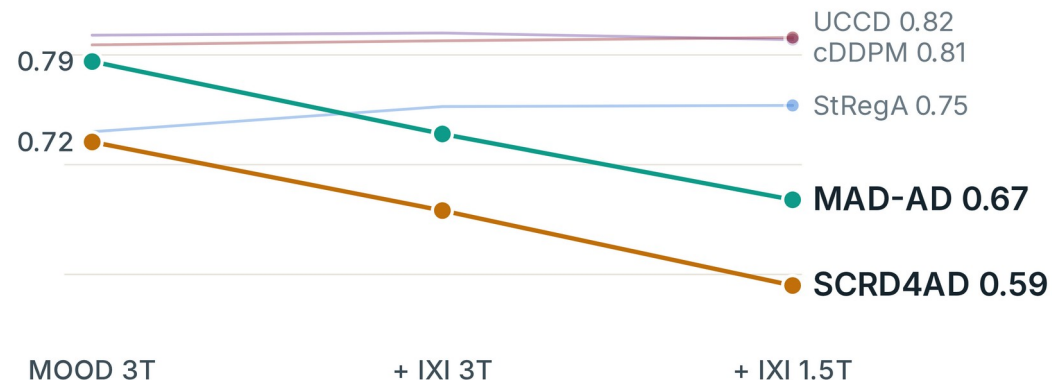
POOLING

More training sites are not always better

-0.13

Slice AUROC lost by MAD-AD as IXI 3T, then IXI 1.5T scans joined MOOD

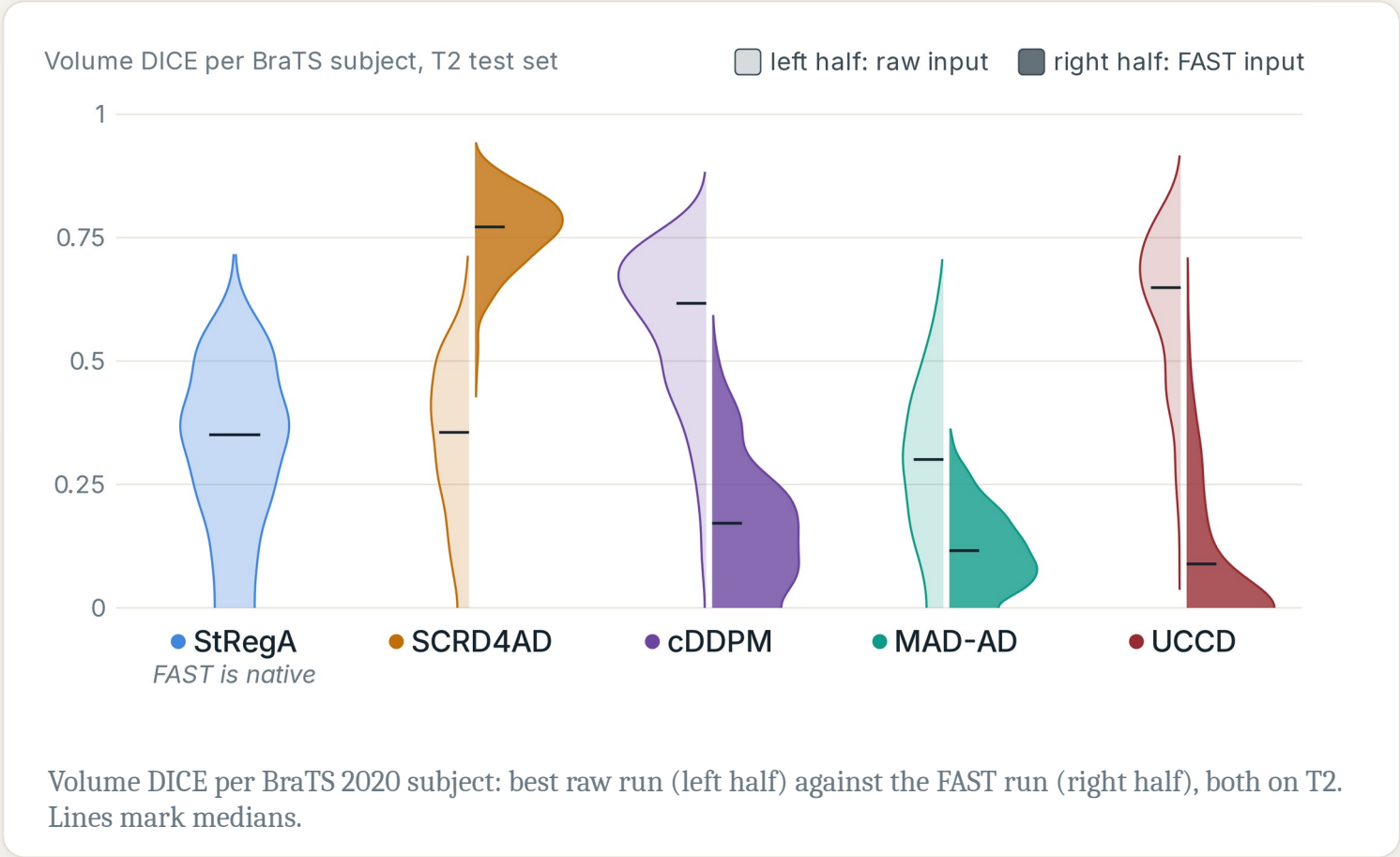
Slice AUROC on BraTS T1 as training sites are added



SCR4AD dropped by a similar amount. The conditional diffusion models, cDDPM and UCCD, held steady.

FAST lifts one model's DICE, sinks three

We gave StRegA's FSL FAST tissue maps, instead of raw intensities, to the other four methods.



IN ONE SENTENCE

The pipeline around a model can matter as much as the model itself.

Contrast, training data and preprocessing each reordered our leaderboard. Report UAD results across a matrix of conditions, not one configuration per method.

Poster PG180

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